Certified Robustness from One Setpoint Step: A Safe Acceptance Test for Model-Free PID Tuning

Pavel Otta[1], Daniel Pachner[1], Jiří Dostál[1] and Vladimír Havlena[2*†]

October 16, 2026

Abstract

Model-free PID tuners improve a loop from closed-loop data but do not guarantee its robustness, and checking the sensitivity peak M_s of a running loop usually needs a model or dedicated experiments. We give a deterministic certificate of M_s from one ordinary setpoint step, the experiment such tuners already perform. In a stable digital loop, the increments of the error after a setpoint step form the impulse response of the sensitivity function; from a finite record we derive computable lower and upper bounds on M_s, with explicit tail, grid and high-frequency terms and one assumption on the unrecorded tail. As an acceptance test, the certificate lets any iterative tuner that applies setpoint steps accept only tunings with $M_s \le M^*$, a uniform back-off of the gains always restores acceptance, and the same record certifies stability and M_s of a proposed tuning before it is applied. We work this out for SPIN, which counts the turns of the control error in phase portraits: the counts bound M_s only on special loops, while certified SPIN provably ends below M^* and safe SPIN never applies a tuning above it. On 19 loops, published SPIN ended above $M^* = 1.7$ on up to 9 and applied tunings above it in 57% of its iterations; safe SPIN, in none.

1 Introduction

Let $L = CP$ be the loop transfer function and $S = 1/(1+L)$ the sensitivity function. Its peak $M_s = \max_\omega |S|$ measures robustness: near 1.4 is safe, near 2 is poor [1]. Model-based tuning rules reach a chosen M_s [1, 2]. Model-free tuners adjust the gains from closed-loop data instead, by iterative feedback tuning [3], extremum seeking [4], virtual reference feedback tuning [5], setpoint overshoot [6] or step-response inspection [7]; they target performance or a calibrated response shape, and none guarantees the M_s of its result. Once a loop runs, its M_s is rarely checked at all [8, 9].

The $\mathscr{H}_\infty$ norm of a system, and hence M_s, can be estimated from data: by power iterations over repeated dedicated experiments [10, 11], by data-driven estimation [12], from limit cycles induced in closed loop [13], or from impulse-response data under a decay assumption, with a one-sided guarantee [14]; margins can be bounded from data [15] or measured by relay tests and closed-loop identification [16, 17]. These methods need dedicated experiments, give estimates or one-sided bounds, and are not built into a tuner. Safe Bayesian optimization tunes controllers without leaving a safe set [18], with probabilistic guarantees under a Gaussian-process model of the cost.

We use the experiment a tuner already performs: a setpoint step. That the error increments after the step are the impulse response of S is elementary (Lemma 1). The contributions are what it enables:

1. a deterministic, two-sided certificate of M_s computable from one finite record (Section 2, Theorem 1);
2. for any iterative model-free tuner, a certified acceptance test (Theorem 2) and a certificate of stability and M_s of a proposed tuning *before* it is applied (Theorem 3), which makes the tuning safe throughout, not only at its end (Section 3);
3. their application to SPIN [7]: why the turn counts of SPIN cannot certify (Lemma 2, Proposition 1), certified SPIN, which ends below M^*, and safe SPIN, which never applies a tuning above it (Section 4, Corollaries 1 and 2).

Section 5 confirms the results in simulation.

2 A Certificate from One Setpoint Step

A digital controller runs with the sampling period T_s. Seen at the samples, the loop it closes is linear and time invariant. Throughout, $L(z)$, $P(z)$, $S = 1/(1+L)$ and $M_s = \max_{\theta\in[0,\pi]} |S(e^{j\theta})|$, with $\theta = \omega T_s$, refer to this digital loop, the loop that actually runs.

Lemma 1 (The step contains S) *Let the loop be stable, have one degree of freedom, and have a strictly proper L (the measurement precedes the control action). Let e_n be the error after a unit setpoint step at $n = 0$, applied from rest and*

*[1]University Centre for Energy Efficient Buildings (UCEEB), Czech Technical University in Prague, Buštěhrad, Czech Republic. Corresponding author: pavel.otta@cvut.cz

†[2]Department of Control Engineering, Faculty of Electrical Engineering, Czech Technical University in Prague, Czech Republic.

without disturbances, so that $e_{-1} = 0$ and $e_0 = 1$, and let $a_n = e_n - e_{n-1}$. Then

$$S(e^{j\theta}) = \sum_{n\geq 0} a_n e^{-jn\theta}. \tag{1}$$

Proof: The z-transform of e is $S(z)/(1-z^{-1})$, so $S = (1-z^{-1})\mathscr{Z}\{e\}$. In a stable loop, a_n decays exponentially, so the series converges on the unit circle. □

A real record is finite, so one assumption covers what it does not show.

Assumption 1 (Long enough record) *The loop is stable, and the record $e_0, \ldots, e_N$ satisfies $\sum_{n>N} |a_n| \leq \tau$ for a known τ.*

In a stable loop, $|a_n| \leq c\rho^n$ with $\rho < 1$, so τ falls geometrically with the record length.

Theorem 1 (Certified bounds from one record) *Let the conditions of Lemma 1 and Assumption 1 hold, and let $\hat{S}(\theta) = \sum_{n=0}^{N} a_n e^{-jn\theta}$. Let G be a uniform grid on $[0, \bar{\theta}]$, $\bar{\theta} \leq \pi$, that has spacing h and contains both ends. For any n_c, let*

$$J = \textstyle\sum_{n=0}^{N} |n - n_c|\,|a_n|, \quad B = \sum_{n=1}^{N-1} |a_{n+1} - a_n| + |a_N| + 2\tau.$$

Then $\underline{M} \leq M_s \leq \overline{M}$, where

$$\underline{M} = \max_{\theta\in G} |\hat{S}(\theta)| - \tau,$$
$$\overline{M} = \max\left\{ \max_{\theta\in G} |\hat{S}(\theta)| + \tfrac{hJ}{2} + \tau,\ 1 + \tfrac{B}{\sin(\bar{\theta}/2)} \right\}, \tag{2}$$

and the second term is omitted if $\bar{\theta} = \pi$. If τ equals the true remaining variation, $N \to \infty$, $h \to 0$ and $\bar{\theta} = \pi$, then both bounds converge to M_s.

Proof: By (1), $|S - \hat{S}| \leq \sum_{n>N} |a_n| \leq \tau$ at every θ, which gives $\underline{M}$. On $[0, \bar{\theta}]$, $|\hat{S}|$ equals $|e^{jn_c\theta}\hat{S}|$, whose derivative is bounded by J. Every θ lies within $h/2$ of a grid point, which gives the first term of $\overline{M}$. On $[\bar{\theta}, \pi]$, let $z = e^{-j\theta}$ and $Z_n = \sum_{m=1}^{n} z^m$, so that $|Z_n| \leq 2/|1-z| = 1/\sin(\theta/2)$. Summation by parts gives $\sum_{n\geq 1} a_n z^n = \sum_{n\geq 1} (a_n - a_{n+1}) Z_n$. Since $\sum_{n\geq N} |a_n - a_{n+1}| \leq |a_N| + 2\tau$, we obtain $|S - a_0| \leq B/\sin(\theta/2)$ with $a_0 = 1$. Convergence follows because, for fixed n_c, J stays bounded as $N \to \infty$, so $hJ \to 0$, and the remaining variation tends to zero. □

Everything in (2) except τ comes from the record, and all grid values follow from one chirp z-transform. The grid term covers the frequencies up to $\bar{\theta}$; above them, B bounds $|S|$ at once, and it is small for a smooth record. Fig. 1 shows both steps on one loop.

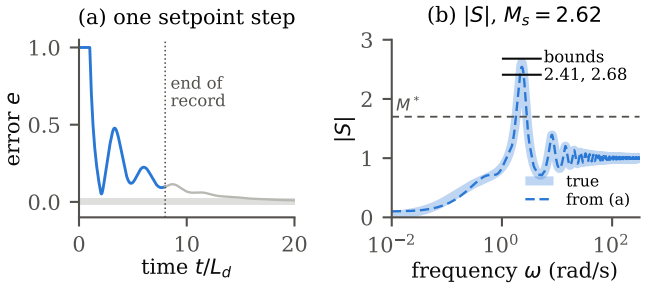


Figure 1: One step certifies M_s. (a) The error after a setpoint step of the slow-integral PI loop of Section 4.2; it never crosses the setpoint by more than 2% (grey band). The record ends at $8L_d$. (b) $|S|$ from the record (Lemma 1) lies on the true $|S|$, and the bounds (2) bracket $M_s = 2.62$, far above M^* (Theorem 1, with $\tau = 0.13$).

3 A Certified Acceptance Test

An iterative tuner applies a controller C, records a setpoint step, and proposes the next controller. *Certified acceptance* keeps the tuner's own rule for proposing, but accepts C only if the upper bound $\overline{M}$ of Theorem 1, computed from the record of C, is at most a target M^*; otherwise it lowers the gains, by the tuner's own rule or by a uniform back-off $C \to (1-\beta)C$. Here C has integral action, $C = C_0 + K_i T_s z^{-1}/(1-z^{-1})$ with C_0 stable, and $\bar{\theta}$ in (2) solves $1 + B/\sin(\bar{\theta}/2) = M^*$ ($\bar{\theta} = \pi$ if no solution), with a grid fine enough that $hJ/2 \leq \eta$.

Theorem 2 (Certified acceptance) *Let the plant be stable, let $K_i P(1) > 0$, and let every tested record meet the conditions of Lemma 1 and Assumption 1.*

(i) Every accepted tuning has $M_s \leq M^*$, whatever rule proposed it.

(ii) For a controller C, let $\mu = \min\{0, \inf_{0<\theta\leq\pi} \mathrm{Re}\, L(e^{j\theta})\}$. For $0 < \lambda < 1/|\mu|$, the loop with λC is stable, and

$$M_s(\lambda C) \leq \frac{1}{1 - \lambda|\mu|}. \tag{3}$$

(iii) If $M^* > 1 + 2\tau + \eta$, then from any such C at most $j^* = \lceil \ln \frac{|\mu|M'}{M'-1} / \ln \frac{1}{1-\beta} \rceil$ uniform back-offs lead to an accepted tuning, where $M' = M^* - 2\tau - \eta$.

Proof: (i) This follows from Theorem 1. (ii) L is bounded on the unit circle except near $z = 1$, where the integral term has a bounded real part, because P is analytic at $z = 1$ and $\mathrm{Re}\, 1/(e^{j\theta} - 1) = -\frac{1}{2}$; so μ is finite. Then $|1 + \lambda L| \geq 1 + \lambda\, \mathrm{Re}\, L \geq 1 - \lambda|\mu| > 0$, which gives (3). The Nyquist curve of λL lies in $\mathrm{Re} \geq -\lambda|\mu| > -1$, also on a small indentation outside $z = 1$, where $K_i P(1) > 0$ keeps $\mathrm{Re}\, L > \mu$; it cannot encircle -1, and L has no poles outside the unit disc, so the loop is stable. (iii) After j back-offs, $\lambda = (1-\beta)^j$, and (3) gives $M_s \leq M'$ for $j \geq j^*$. In (2), the grid term is then at most $M_s + 2\tau + \eta \leq M^*$, and the second term is at most M^* by the choice of $\bar{\theta}$; so the test passes. □

Acceptance tests a tuning after it has been applied. The same record also certifies any other controller *before* it is applied, so a tuner can avoid applying a fragile tuning at all.

Theorem 3 (Certify before applying) *Let the record of C meet the conditions of Lemma 1 and Assumption 1, with C as in Theorem 2, and let C' be such a controller with the same T_s, with $K_i' > 0$ and $\Delta = C'/C - 1$ bounded on the unit circle. Let $\hat{T} = 1 - \hat{S}$. Then $S' = S/(1+T\Delta)$ exactly, with $T = 1 - S$, and if, for all $\theta \in (0, \pi]$,*

$$d(\theta) = \min_{0 \le \lambda \le 1} |1 + \lambda \hat{T}\Delta| - \tau|\Delta| > 0, \tag{4}$$

then C' stabilizes the plant, and

$$M_s(C') \le \sup_{0 < \theta \le \pi} \frac{|\hat{S}| + \tau}{|1 + \hat{T}\Delta| - \tau|\Delta|}. \tag{5}$$

Proof: $L' = L(1+\Delta)$ gives $1 + L' = (1+L)(1+T\Delta)$. Let $C_\lambda = C(1+\lambda\Delta) = (1-\lambda)C + \lambda C'$, $\lambda \in [0,1]$. Its integral gain is positive and its other poles are those of C_0 and C_0', inside the unit disc, so L_λ has no poles outside the unit disc and is large on a small indentation around $z = 1$. Since $|T - \hat{T}| = |S - \hat{S}| \le \tau$, $|1 + \lambda T\Delta| \ge d > 0$, so $1 + L_\lambda \ne 0$ on the indented circle for every λ. The winding number of $1 + L_\lambda$ about 0 is thus constant in λ, and at $\lambda = 0$ the loop is stable. The bound follows from $|S| \le |\hat{S}| + \tau$ and $|1 + T\Delta| \ge |1 + \hat{T}\Delta| - \tau|\Delta|$. □

Like (2), (4) and (5) are checked on a grid with Lipschitz margins; above $\bar{\theta}$, the bound $|S - 1| \le r = B/\sin(\theta/2)$ from the proof of Theorem 1 replaces the record, and (4) holds if $r|\Delta| < 1$, with $|S'| \le (1+r)/(1 - r|\Delta|)$. As $\tau \to 0$, (5) becomes the exact prediction $\max |\hat{S}/(1 + \hat{T}\Delta)|$, and a back-off is the case $\Delta = \lambda - 1$.

4 Application to SPIN

4.1 The SPIN method

SPIN [7] tunes an I, PI or PID controller with the gains K_i, K_p, K_d. In each iteration, it applies one setpoint step, records the error e, and forms three phase portraits,

$$\Gamma_0 : (e, \textstyle\sum e), \quad \Gamma_1 : (\Delta e, e), \quad \Gamma_2 : (\Delta^2 e, \Delta e), \tag{6}$$

where $\sum e$ is the running sum of e and Δ the difference of consecutive samples. They correspond to the frequency bands of K_i, K_p and K_d, which we call bands 0, 1 and 2. Each axis is divided by its peak, and the count N_k is the number of turns that Γ_k makes around its end point before the error stays within $\pm\delta$ (2% of the step). Algorithm 1 states the rule.[1]

4.2 Why the turn counts cannot certify

In each portrait (6), the second coordinate q is the running sum of the first coordinate p. The count is the net counterclockwise angle around the end point, placed at the origin, divided by 2π; we state the results in continuous time, where $\dot{q} = p$. For Γ_0, the zeros of $p = e$ are the setpoint crossings.

[1] An online simulator: `https://robopid-simulator.uceeb.cvut.cz`

Algorithm 1: one iteration of SPIN [7] and, with lines 7a–7c in bold instead of line 7, of certified SPIN (Section 4.3)

```
 1: apply a setpoint step and record the error e
 2: if the record does not settle then
 3:     divide K_i, K_p, K_d by 2, 4, 8                (retreat)
 4: else
 5:     count the turns N_0, N_1, N_2 of Γ_0, Γ_1, Γ_2
 6:     if N_k ≤ N̄_k for every band k then
 7:         raise all gains by one increment           (accept)
7a:     compute M̄ from the record (Theorem 1)
7b:     if M̄ ≤ M* then raise all gains                 (accept)
7c:     else lower the gains: the band move that
            Thm. 3 predicts to lower M_s (highest
            such band, at most n_max in a row), else a
            uniform back-off                           (reject)
 8:     else
 9:         k ← lowest band with N_k > N̄_k
10:         lower the gain of band k by one increment and
            raise the gains of the bands below k by one
```

An increment scales a gain by $1/(1-\beta)$. Defaults: $\bar{N} = (0.5, 0.75, 1.0)$, $\delta = 0.02$, $\beta = 0.1$; gains stay within $[0.001, 10]$ times their initial values (the gain box); the result is the last accepted tuning. Certified SPIN: $\bar{\theta}$ and h as in Section 3; $M^* = 1.7$, $\tau = 10^{-3}$, $\eta = 2 \cdot 10^{-3}$, $n_{\max} = 10$.

Lemma 2 (Half turns) *Let $\dot{q} = p$ with p continuous, let $p(t_1) = p(t_2) = 0$ with $q(t_1)q(t_2) < 0$, and let p not change sign on (t_1, t_2). Then, on $[t_1, t_2]$, the curve (p, q) turns counterclockwise around the origin by exactly half a turn, and so does its image under any linear map with a positive determinant, such as the per-axis normalization of SPIN.*

Proof: On (t_1, t_2), p keeps one sign, so the curve stays in one closed half-plane bounded by the q-axis, and q moves monotonically from one ray of the q-axis to the opposite ray; it avoids the origin, since $q = 0$ only where $p \ne 0$. The angle thus changes by exactly π, and counterclockwise, since q increases where $p > 0$ and decreases where $p < 0$. A linear map with a positive determinant maps half-planes to half-planes and opposite rays to opposite rays, and keeps the orientation. □

So each setpoint crossing adds half a turn to N_0, and how many crossings a count sees depends on how many rise above δ. Whether crossings reveal M_s depends on the loop. Lambda, IMC and SIMC tunings cancel the plant lags [2], e.g., PI with $T_i = T$ on $Ke^{-sL_d}/(Ts+1)$; the loop is then

$$L(s) = \frac{k\, e^{-sL_d}}{sL_d}, \qquad k = \frac{KK_cL_d}{T_i}. \tag{7}$$

Proposition 1 (Crossings on lag-cancelled loops) *The loop (7) is stable for $0 < k < \pi/2$, and its M_s strictly increases with k, as given by*

$$k = x\sin x - x^2\cos x, \qquad \frac{1}{M_s^2} = 1 - \frac{2k\sin x}{x} + \frac{k^2}{x^2}, \tag{8}$$

with $x \in (0, \pi/2)$ the normalized peak frequency. The error after a setpoint step never crosses the setpoint if and only if $k \le 1/\mathrm{e}$, that is, $M_s \le 1.394$; otherwise it crosses infinitely often.

Proof: The phase of L is $-\pi/2 - x$ at $x = \omega L_d$; it reaches $-\pi$ first at $x = \pi/2$, where $|L| = 2k/\pi < 1$, and $|L| = k/x$ decreases, so the loop is stable. Let $h(x) = |1+L|^2 = 1 - 2k\sin x/x + k^2/x^2$. Then $h'(x) = 2k(\varphi(x) - k)/x^3$ with $\varphi(x) = x\sin x - x^2\cos x$ increasing on $(0,\pi)$ and $\varphi(\pi/2) > k$, so h has one minimum $x^* \in (0,\pi/2)$ on $(0,\pi)$; on $[\pi, 2\pi]$, $h \ge 1$, and for $x \ge 2\pi$, $h \ge (1 - k/x)^2 > h(\pi/2)$. This gives (8), and by the envelope theorem $\frac{d}{dk}h(x^*) = -2\cos x^* < 0$. The error obeys $\dot{e}(t) = -(k/L_d)e(t - L_d)$ with $e = 1$ on $[0, L_d]$; it stays positive if and only if the characteristic equation $s + (k/L_d)e^{-sL_d} = 0$ has a real root, that is, if $k \le 1/\mathrm{e}$, and otherwise all solutions oscillate [19]. □

At $\delta = 0.02$ the counts of the loop (7) therefore form a staircase in M_s (Fig. 2(a)). With n crossings above δ, the computed N_0 and N_1 lie within a quarter turn above $n/2$, and N_2 is $n/2 - 1/4$ within 0.01: on this ideal loop $\dot{e}$ jumps at $t = L_d$, so Γ_2 starts with a quarter turn clockwise. The published limits lie in the gaps: $\bar{N}_0 = 1/2$, $\bar{N}_1 = 3/4$ and $\bar{N}_2 = 1$ allow 0, 1 and 2 crossings above δ, that is, $M_s \le 1.534$, 1.842 and 2.210, where the $(n+1)$th extremum equals δ. Each band allows one crossing more than the band below, which is why the limits grow; Γ_0 binds, so on these loops SPIN stops at $M_s \le 1.534$.

On general loops, this guarantee is lost. The counts leave the staircase: in the 3800 count decisions of Section 5, N_0 and N_2 fell between $n/2 + 1/4$ and $(n+1)/2$ in 22 and 67% of them. And crossings do not bound M_s: a first-order plant with $T = 0.3L_d$ under PI with $T_i = 8T$ and a gain margin of 1.624 has $M_s = 2.62$, yet its error never crosses the setpoint by more than 2% (Fig. 1(a)), while a controller zero can cause crossings at a low M_s. The counts can shape a tuning, but only a certificate can certify it.

4.3 Certified SPIN

Certified SPIN accepts a tuning that the counts accept only if the upper bound $\overline{M}$ of Theorem 1, computed from the same record, is at most a target M^*. In Algorithm 1, it replaces line 7 by the bold lines 7a–7c: the counts shape the tuning, and the certificate decides whether it is accepted. A rejected tuning is lowered by the band move that the prediction of Theorem 3 expects to lower M_s, as in line 10, or by a uniform back-off, the reverse of an acceptance.

Corollary 1 (Certified SPIN) *Under the conditions of Theorem 2, the result of certified SPIN has $M_s \le M^*$, whatever the limits $\bar{N}_k$. If no gain is clipped by the gain box, every run of rejections in line 7b ends in an acceptance or in a move of Algorithm 1 (line 3 or 10), after at most $n_{\max}$ band moves and at most j^* back-offs in each stretch of back-offs.*

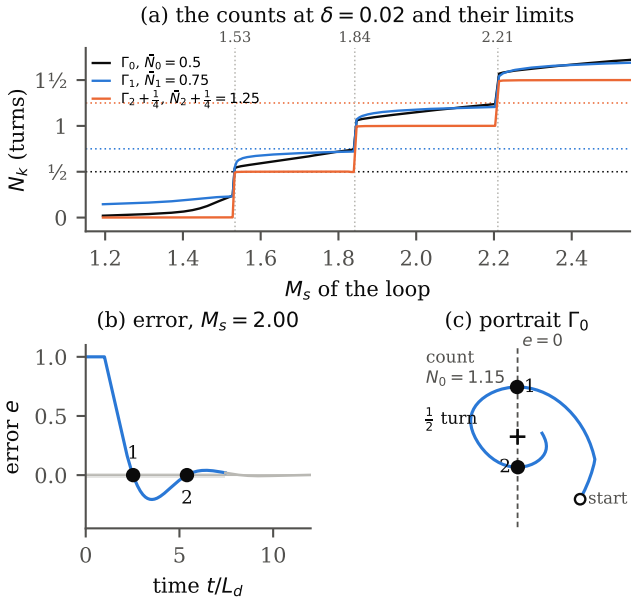


Figure 2: The lag-cancelled loop (7), SPIN's counts. (a) At $\delta = 0.02$, each count steps over its published limit (dotted) at $M_s = 1.53$, 1.84, 2.21; N_2 and its limit are drawn a quarter turn up (see text). (b) A record with $M_s = 2.00$ and two crossings. (c) Its portrait Γ_0 turns by exactly half a turn between them (Lemma 2); the radius is compressed, the angle is not.

Proof: Line 7c allows at most $n_{\max}$ band moves in a row; the rest follows from Theorem 2(i) and (iii), unless the counts reject first. □

So the test never traps SPIN. It does not stop SPIN from *applying* fragile tunings: each is tested before it is rejected. *Safe SPIN* certifies every move before it is applied. It applies the move of Algorithm 1 only if Theorem 3 certifies it with a bound (5) at most M^*; otherwise the first so certified of the band moves $k = 2, 1, 0$ and the uniform back-off; otherwise it keeps the current tuning. Until line 7b first accepts, it backs off uniformly; a tuning certified ahead that line 7b does not accept is replaced by the last accepted one.

Corollary 2 (Safe SPIN) *Under the conditions of Theorem 2, from the first accepted tuning on, every tuning that safe SPIN applies stabilizes the plant and has $M_s \le M^*$, whatever the limits $\bar{N}_k$.*

Proof: Moves are proposed only from accepted tunings, so each applied tuning is certified by Theorem 3 from the record of an accepted tuning, or was accepted itself (Theorem 2(i)). □

5 Simulations

Setup We compare published and certified SPIN. Both run 200 iterations with a derivative filter ratio of 10 and

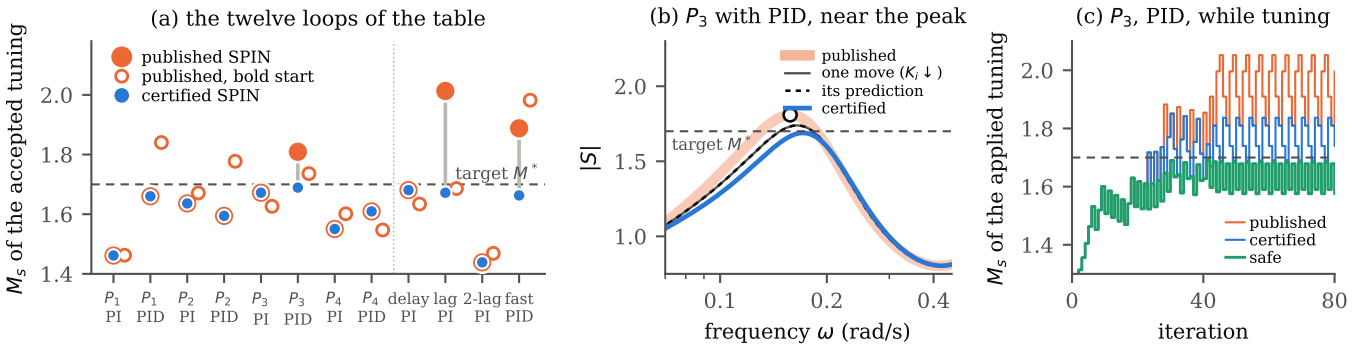


Figure 3: Published vs. certified SPIN, $M^* = 1.7$. (a) The loops of Table 1. (b) P_3 with PID: published tuning ($\overline{M} = 1.81$, ∘), one band move lowering K_i ($M_s = 1.739$, as Theorem 3 predicts), certified tuning (1.689). (c) P_3 with PID: M_s of the tuning applied in each iteration (fixed T_s); safe SPIN stays below M^* throughout.

Table 1: M_s of the digital loop for the most aggressive tuning that published and certified SPIN accept, from the cautious and the bold start, and the change of the load-step IAE (cautious start; –: tuning unchanged).

		published		certified		
		caut.	bold	caut.	bold	ΔIAE
P_1	PI	1.46	1.46	1.46	1.46	–
	PID	1.66	**1.84**	1.66	1.61	–
P_2	PI	1.64	1.67	1.64	1.67	–
	PID	1.59	**1.78**	1.59	1.66	–
P_3	PI	1.67	1.63	1.67	1.63	–
	PID	**1.81**	**1.74**	1.69	1.60	+23%
P_4	PI	1.55	1.60	1.55	1.60	–
	PID	1.61	1.55	1.61	1.55	–
delay-dom.	PI	1.68	1.63	1.68	1.63	–
lag-dom.	PI	**2.01**	1.69	1.67	1.69	+52%
2 lags	PI	1.44	1.47	1.44	1.47	–
fast	PID	**1.89**	**1.98**	1.66	1.69	0%

the gain box of [7]. Since the box ties the result to the start, every loop runs from SIMC on a half-rule model with τ_c three times its dead time (cautious start) and from the same tuning with all gains doubled (bold start). The 19 test loops are (1) the four plants of [7], $P_1 = e^{-s}/((10s+1)(s+1)^3)$, $P_2 = 1.25e^{-8s}/(5s+1)^4$, $P_3 = e^{-10s}/((2s+1)(s+1)^3)$ and $P_4 = e^{-4s}/(8s+1)^6$, with PI and with PID; (2) four plants that place the oscillation in a chosen band: the delay-dominant $e^{-5s}/(s+1)$ and the lag-dominant $e^{-s}/(10s+1)$ and $e^{-0.5s}/((10s+1)(s+1))$ with PI, and the fast $e^{-0.1s}/((s+1)(0.5s+1))$ with PID; (3) the family $e^{-L_d s}/((s+1)(0.5s+1)(0.25s+1))$ with PID and $L_d \in \{0.25, 0.5, 1, \ldots, 16\}$ (dead-time share 0.125 to 0.90). P, I and D act on the error. Let T_a be the dead time plus the lag time constants and T_f the derivative filter time constant; the controller runs at $T_s = \min(T_a/100, T_f/20)$ for PID and $T_a/1000$ for PI, with exactly discretized lags and the delay rounded to whole samples. Each record covers $15\,T_a$. All values of M_s are those of this digital loop, computed from its exact L. Because SPIN ends in a cycle between a few tunings, we report the most aggressive tuning accepted in the last 60 iterations.

The certificate (Section 2) For all 146 tunings accepted below, a four times longer record showed a remaining variation $\leq 2.8 \cdot 10^{-5}$, so Assumption 1 held with $\tau = 10^{-3}$; (1) held up to this variation, and every M_s lay within its bounds, with $\overline{M} - M_s \leq \eta + \tau + 2.8 \cdot 10^{-5}$. Short records give wider bounds that still contain M_s (Fig. 1(b)).

The counts (Section 4.2) On the lag-cancelled loop, Γ_0 turned by half a turn between crossings within 10^{-6}, also after a linear map, and (8) matched a direct search for M_s to six digits.

Certified SPIN (Section 4.3; Table 1, Fig. 3)

- *Safety.* Published SPIN ended above $M^* = 1.7$ on 9 of the 19 loops from the cautious start (up to 2.01), on 8 from the bold start (up to 2.09) and, without the gain box, on 12 of 18 (up to 2.05); certified SPIN on none (at most 1.69), as Theorem 2(i) guarantees.
- *No trapping.* (3) held along each back-off, predicted and true M_s after every band move agreed within 10^{-7}, and runs of rejections had at most 4 band moves; every loop kept accepting, except $L_d = 16$ without the box, where the counts lower K_d every round under either rule.
- *Cost.* The load-step IAE rose by 7.8% on average (6.4% from the bold start).

Safety during tuning (Corollary 2, Fig. 3(c)) Theorem 3 needs one T_s, so we reran all three variants with T_s fixed at its value for the start tuning (published SPIN then ended above M^* on 9 and 7 loops). In the 7600 iterations of the 38 runs, published SPIN applied a tuning with $M_s > M^*$ in 4371 (up to $M_s = 2.38$), certified SPIN, which must apply a tuning to reject it, in 3398 (up to 1.98), and safe SPIN in none: all its tunings were stable, with $M_s \leq 1.696$. Every start tuning was

accepted. Safe SPIN ended with $M_s \leq 1.694$ and the same median load-step IAE as certified SPIN, but up to twice as large on the fast PID loop. On 456 random moves (gains scaled by up to 8) from accepted tunings, (4) held for 339, all stable, and failed for 117, all unstable; (5) exceeded M_s by a median of 0.011 and certified 116 of the 118 moves with $M_s \leq M^*$.

Limit values under the certificate With every limit $1/2$, certified SPIN still stayed below M^* (at most 1.68), but the counts did the rejecting (the certificate 14 and 8 times in the last 60 iterations, against 151 and 172): 2 and 3 loops accepted no tuning, and the load-step IAE rose by a median of 11 and 16%, up to sixteenfold (fast PID, $M_s = 1.21$). Raising $\bar{N}_1$ to 1 changed nothing, since N_1 never lay in $(3/4, 1]$. Under the certificate, the limits set performance, not robustness.

6 Conclusion

The setpoint step that a model-free tuner applies anyway certifies the robustness of the loop it tests: one finite record gives two-sided bounds on M_s without a model and without a dedicated experiment, and so does every logged setpoint change made from rest. As an acceptance test, the certificate makes any iterative tuner accept only tunings with $M_s \leq M^*$, a uniform back-off always restores acceptance, and certifying each move before it is applied keeps every applied tuning below M^*. For SPIN, the turn counts measure setpoint crossings and bound M_s only on special loops; with the certificate they keep shaping the tuning, and the certificate decides what is accepted. The theorems assume a linear loop and a long enough record; noise and real loops are left to a follow-up. Scripts: `https://github.com/ottapav/robopid-simulator`.

References

[1] K. J. Åström and T. Hägglund, "Revisiting the Ziegler–Nichols step response method for PID control," *J. Process Control*, vol. 14, no. 6, pp. 635–650, 2004.

[2] S. Skogestad, "Simple analytic rules for model reduction and PID controller tuning," *J. Process Control*, vol. 13, no. 4, pp. 291–309, 2003.

[3] H. Hjalmarsson, M. Gevers, S. Gunnarsson, and O. Lequin, "Iterative feedback tuning: theory and applications," *IEEE Control Syst. Mag.*, vol. 18, no. 4, pp. 26–41, 1998.

[4] N. J. Killingsworth and M. Krstić, "PID tuning using extremum seeking: online, model-free performance optimization," *IEEE Control Syst. Mag.*, vol. 26, no. 1, pp. 70–79, 2006.

[5] M. C. Campi, A. Lecchini, and S. M. Savaresi, "Virtual reference feedback tuning: a direct method for the design of feedback controllers," *Automatica*, vol. 38, no. 8, pp. 1337–1346, 2002.

[6] M. Shamsuzzoha and S. Skogestad, "The setpoint overshoot method: a simple and fast closed-loop approach for PID tuning," *J. Process Control*, vol. 20, no. 10, pp. 1220–1234, 2010.

[7] D. Pachner, P. Otta, J. Dostál, and V. Havlena, "Model-free PID tuning by step-response inspection," submitted to *J. Process Control*, 2026.

[8] L. Desborough and R. Miller, "Increasing customer value of industrial control performance monitoring—Honeywell's experience," in *Chemical Process Control VI*, AIChE Symp. Ser. no. 326, vol. 98, 2002, pp. 169–188.

[9] M. Bauer, A. Horch, L. Xie, M. Jelali, and N. Thornhill, "The current state of control loop performance monitoring—a survey of application in industry," *J. Process Control*, vol. 38, pp. 1–10, 2016.

[10] B. Wahlberg, M. Barenthin, and H. Hjalmarsson, "Non-parametric methods for L_2-gain estimation using iterative experiments," *Automatica*, vol. 46, no. 8, pp. 1376–1381, 2010.

[11] C. R. Rojas, T. Oomen, H. Hjalmarsson, and B. Wahlberg, "Analyzing iterations in identification with application to nonparametric $\mathscr{H}_\infty$-norm estimation," *Automatica*, vol. 48, no. 11, pp. 2776–2790, 2012.

[12] K. van Heusden, A. Karimi, and D. Bonvin, "Data-driven estimation of the infinity norm of a dynamical system," in *Proc. 46th IEEE Conf. Decision and Control*, New Orleans, 2007.

[13] D. Garcia, A. Karimi, and R. Longchamp, "Infinity norm measurement of sensitivity function based on limit cycles in a closed-loop experiment," in *Proc. European Control Conf.*, Cambridge, UK, 2003, pp. 2073–2078.

[14] S. Tu, R. Boczar, and B. Recht, "On the approximation of Toeplitz operators for nonparametric $\mathscr{H}_\infty$-norm estimation," arXiv:1709.10203, 2017.

[15] K. Isoshima, M. Tanemura, and Y. Chida, "Data-driven estimation of the lower bounds of gain and phase margins," *Automatica*, vol. 153, art. 111008, 2023.

[16] Q.-G. Wang, C. C. Hang, and Q. Bi, "Process frequency response estimation from relay feedback," *Control Eng. Pract.*, vol. 5, no. 9, pp. 1293–1302, 1997.

[17] P. M. J. Van den Hof and R. J. P. Schrama, "An indirect method for transfer function estimation from closed loop data," *Automatica*, vol. 29, no. 6, pp. 1523–1527, 1993.

[18] F. Berkenkamp, A. P. Schoellig, and A. Krause, "Safe controller optimization for quadrotors with Gaussian processes," in *Proc. IEEE Int. Conf. Robotics and Automation*, Stockholm, 2016, pp. 491–496.

[19] I. Győri and G. Ladas, *Oscillation Theory of Delay Differential Equations with Applications*. Oxford: Clarendon Press, 1991.